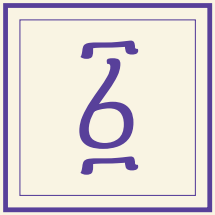
Business cases

Bandits

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026



Section 1

Spotify’s home page

Spotify's home page: music or podcasts?

The home page shows shelves of music and podcasts. A common rule gives users the mix of their past listening.

But tastes can change with the hour, the day of the week, even the device.

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

Spotify's home page: music or podcasts?

The home page shows shelves of music and podcasts. A common rule gives users the mix of their past listening.

But tastes can change with the hour, the day of the week, even the device.

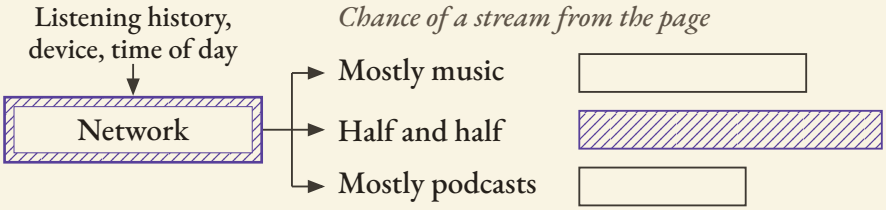
Over 600 million

users of the home page

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

Epsilon-greedy on the home page

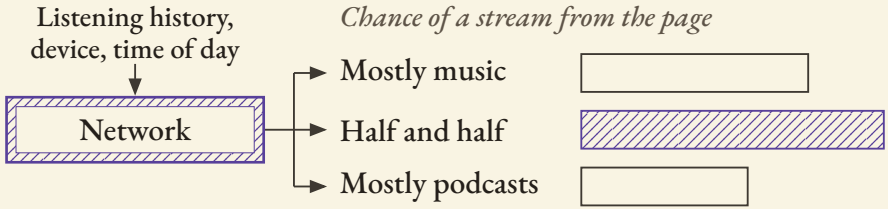
The mirror counted taps. Here a network estimates the chances.



Invented mixes and estimates. Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

Epsilon-greedy on the home page

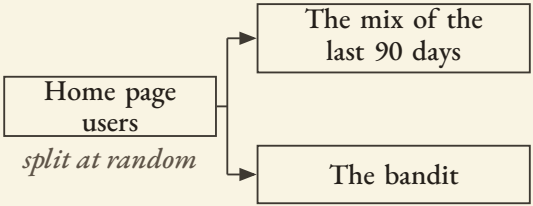
The mirror counted taps. Here a network estimates the chances.



98.5% of the pages: the highest estimate.
1.5%: a random mix near the usual one.

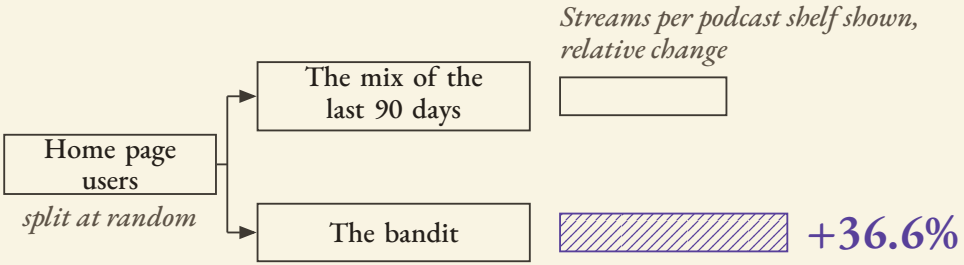
Invented mixes and estimates. Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

The value: an A/B test



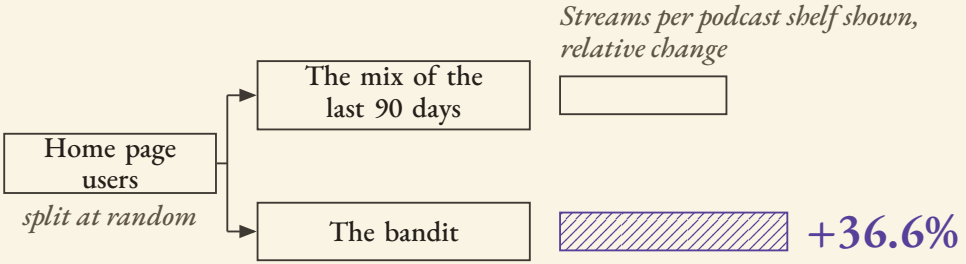
Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

The value: an A/B test



Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

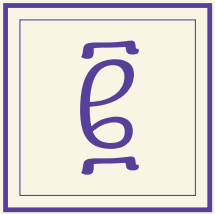
The value: an A/B test



In production as of March 2025.

+1.28% *minutes listened on Spotify, relative change*

Feijer and others (Spotify), CONSEQUENCES workshop at RecSys 2025.

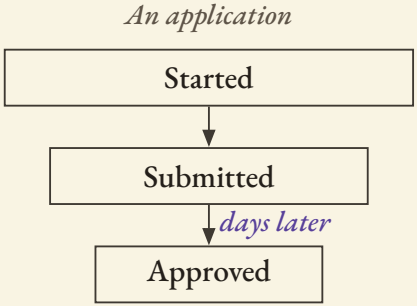


Section 2

Amazon Payments

Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

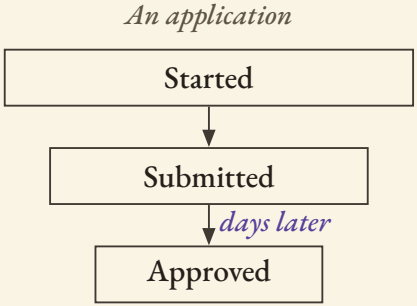


John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

Amazon Payments: which page for which visitor?

A page invites customers to apply for an Amazon Payments product. It shows one image and one tagline, from many possible pairs.

A page that wins the first step can lose the last.



John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*
Pair A	6	3	1
Pair B	4	4	4
Pair C	3	2	3

The first step alone would choose pair A.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

One UCB score for each step

Each score is an estimate plus a bonus, as in the mirror.

For one visitor	*Started*	*Submitted*	*Approved*	*Sum*
Pair A	6	3	1	10
Pair B	4	4	4	12
Pair C	3	2	3	8

The first step alone would choose pair A.
Amazon shows the highest sum: pair B.

Invented scores. John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers



Approvals rose by a high single-digit percentage.
relative change so far, not reported as significant

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers		Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant*
Another group		No pair it tried beat the fixed page. Approvals fell. *statistically significant*

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.

The value: an A/B test of 7 weeks

The bandit against the fixed page:

One group of customers		Approvals rose by a high single-digit percentage. *relative change so far, not reported as significant*
Another group		No pair it tried beat the fixed page. Approvals fell. *statistically significant*

Amazon: “Thorough exploration can confirm that a content pool contains no winner.”

John and Gozluklu (Amazon), AWS Machine Learning Blog, 2026.